

Data Quality & Semantic Profiling Report

Collection: test_collection | Generated At: 2026-06-07 19:45:52 | Total Documents: 3

1. Data Quality Score

Grade D

47.0 / 100

Semantic profile indicates data quality stands at 47.0% with grade D. Details of quality checks are cataloged below

2. Quality Violations & Findings

Field Path	Rule Viol	Sev	Rate %	Examples	Actionable Suggestion
birth_date	FutureDate	CRITICAL	33.3% (1)	20*****07	Fix timestamp generation logic; dates in 'birth_date' cannot be in the future.
created_at	AncientDate	MEDIUM	33.3% (1)	19*****9Z	Date in 'created_at' is before year 2000, verify context accuracy.
description	EmptyString	INFO	33.3% (1)		Store null or remove empty space elements for field 'description'.
email	MalformedEmail	HIGH	33.3% (1)	in*****ss	Format field 'email' to strictly follow RFC 5322 structure.
paid	FakeBoolean	MEDIUM	33.3% (1)	!***	Cast phone/string boolean values (e.g. 0/1, 'true') to valid bool in field 'paid'.
phone	MalformedPhone	HIGH	66.7% (2)	1**, 55***cd	Sanitize phone field 'phone' to contain only digits and standard separators.
price	NegativeValue	HIGH	33.3% (1)	-***	Enforce value validations to block negative entries in field 'price'.
price	ZeroValue	MEDIUM	33.3% (1)	0**	Verify business logic: field 'price' should not be zero.
total	ZeroValue	MEDIUM	33.3% (1)	0**	Verify business logic: field 'total' should not be zero.
age, birth_date	AgeVsBirthdate	HIGH	33.3% (1)	Age: 20, Birth: 1990-05-15	Validate and align age values with birth_date birth year.
status, delivery_date	StatusVsDate	HIGH	33.3% (1)	Status: delivered, Date: None	Add delivered timestamps to track delivered packages.
items, total	InconsistentTotal	HIGH	33.3% (1)	Total: 100.0, Expected: 50.00	Align document total field with the aggregate sum of item prices and quantities.

3. Field Semantics & Statistics Profiles

Field Path	Dominant Type	Null Rate	Unique Rate	Min / Max / Avg	Semantic Inferred	Top Values
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age	float	0.0%	66.7%	Min: 20.0 Max: 36.0 Avg: 28.00	TEXT	36.0, 20.0
birth_date	str	0.0%	66.7%	Min: 1990-05-15 Max: 2026-07-07 Avg: -	DATE	1990-05-15, 2026-07-07
comment	float	0.0%	33.3%	Min: TODO: fix this value Max: TODO: fix this value Avg: -	TEXT	TODO: fix this value
created_at	str	0.0%	66.7%	Min: 1999-12-31T23:5 9:59Z Max: 2026-01-01T00: 00:00Z Avg: -	DATE	2026-01-01T00: 00:00Z, 1999-12 -31T23:59:59Z
delivery_date	float	0.0%	33.3%	Min: 2026-06-01T12:0 0:00Z Max: 2026-06-01T12: 00:00Z Avg: -	DATE	2026-06-01T12: 00:00Z
description	float	0.0%	33.3%	Min: Max: Avg: -	TEXT	
email	str	0.0%	100.0%	Min: admin@test.com Max: john.doe@exam ple.com Avg: -	EMAIL	john.doe@exam ple.com, invalid- email-address, admin@test.co m
id	int	0.0%	100.0%	Min: 1.0 Max: 3.0 Avg: 2.00	ID	1, 2, 3
items	list	0.0%	100.0%	Min: [] Max: None Avg: -	TEXT	[{'price': 19.99, 'qty': 2}], [], [{'price': 10.0, 'qty': 5}]
items.price	float	0.0%	66.7%	Min: 10.0 Max: 19.99 Avg: 14.99	PRICE	19.99, 10.0
items.qty	float	0.0%	66.7%	Min: 2.0 Max: 5.0 Avg: 3.50	TEXT	2.0, 5.0
paid	bool	0.0%	100.0%	Min: 0.0 Max: 1.0 Avg: 0.50	ID	True, true, False
phone	str	0.0%	100.0%	Min: 123.0 Max: 123.0 Avg: 123.00	PHONE	+1-555-0199, 123, 555-abcd
price	float	0.0%	100.0%	Min: -5.0 Max: 19.99 Avg: 5.00	PRICE	19.99, -5.0, 0.0
status	str	0.0%	66.7%	Min: delivered Max: pending Avg: -	TEXT	delivered, pending
total	float	0.0%	100.0%	Min: 0.0 Max: 100.0 Avg: 46.66	PRICE	39.98, 0.0, 100.0